

Arkaprabha Banerjee | AI/ML Engineer & Full-Stack Developer

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Summary

Undergraduate in Computer Science (Data Science) with a CGPA 9.1 (last sem), 8.57 overall and demonstrated expertise in end-to-end development of AI/ML systems. Delivered production-grade applications in NLP, computer vision, and retrieval-augmented generation (RAG). Proven ability to optimize system performance, reduce latency, and deploy scalable full-stack solutions for real-world impact.

Education

Heritage Institute of Technology

B.Tech. in Computer Science (Data Science), CGPA: 9.1 2022–2026
Data Structures & Algorithms, Database Management Systems, Software Engineering, Machine Learning, Computer Vision, Web Development

D.A.V Public School

Class X (88.8%) and Class XII (90.6%) 2020–2022

Experience

TCS

Data Science Intern Aug–Oct 2025
Enterprise analytics, ML workflows, and large-scale data processing using Python .

Noobuild Community

DSA Mentor Sep 2024–Jun 2025
Mentoring early developers in DSA via a 10-month structured program.

Projects & Achievements

KRISHAK – Agri AI Platform

2025
Built India's first SMS-based AI advisory for feature-phone farmers (FastAPI, Twilio, PostgreSQL); achieved 3s response across 36+ states via query optimization.

Designed RAG pipeline with Qdrant, Sentence Transformers, and dual LLMs (Gemini Pro, Qwen)—85%+ recommendation accuracy. Reduced farmer decision latency by 70% and communication costs by 80% using AI-driven climate and carbon credit insights.

Winner, Status Code 2 Hackathon.

RAG Multi-Agent Q&A Assistant

2025
Developed multi-agent retrieval system using LangChain, FAISS, Qwen (Groq), and E5 embeddings for semantic search. Streamlit UI cut average query resolution time by 30%.

Technical Skills

Languages: Python, C++ , Java , JavaScript, SQL

Backend: FastAPI, Django, Flask | REST APIs, Auth, Rate Limiting, Groq

Databases: PostgreSQL, MySQL, SQLite | Schema Design, Query Optimization

ML/AI: PyTorch, TensorFlow, Scikit-learn | NLP, Computer Vision, MLOps

Data Tools: Pandas, NumPy, Matplotlib